

LAB EXPERIMENT RECORD

Set 6: Product Sales Analysis

Department	AIML
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To analyze monthly product sales using a grouped bar chart, stacked area chart, and data table, and to build an interactive dashboard using R and Shiny.

Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

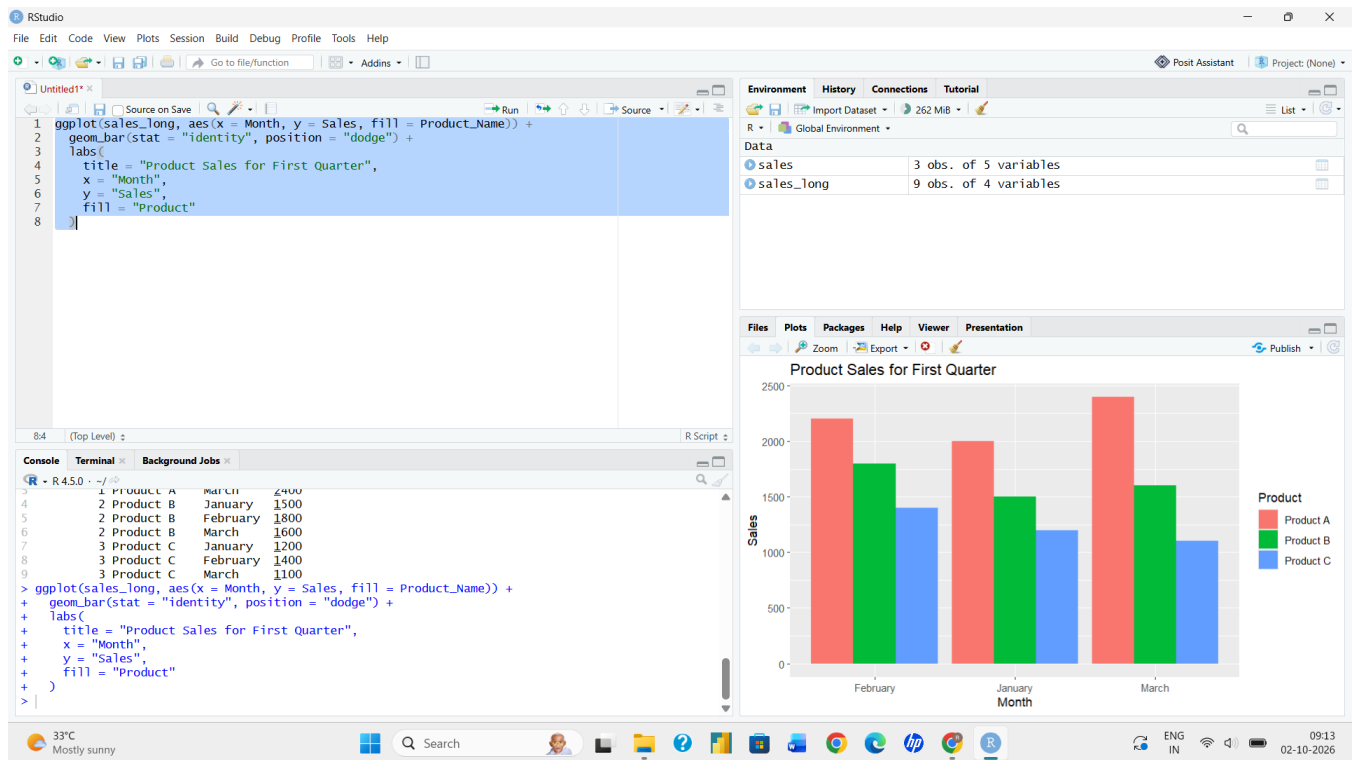
PROGRAM 1: Grouped Bar Chart of Quarterly Sales

Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

```
# Dataset
product <- c("Product A", "Product B", "Product C")
jan <- c(2000, 1500, 1200)
feb <- c(2200, 1800, 1400)
mar <- c(2400, 1600, 1100)

# Matrix: rows = months, columns = products
sales <- rbind(jan, feb, mar)
colnames(sales) <- product
rownames(sales) <- c("January", "February", "March")

# Grouped Bar Chart
barplot(sales,
  beside = TRUE,
  xlab = "Product",
  ylab = "Sales",
  main = "Product Sales in the First Quarter",
  col = c("skyblue", "orange", "seagreen"),
  legend.text = rownames(sales),
  args.legend = list(x = "topright"),
  ylim = c(0, 3000))
```



Output 1: Grouped Bar Chart of Product Sales

Observation: Product A has the highest sales in every month and grows steadily from 2000 to 2400. Product B peaks in February (1800), and Product C also peaks in February (1400) before falling to 1100 in March.

PROGRAM 2: Stacked Area Chart of Overall Sales Trend

Generate a stacked area chart to represent the overall sales trend for all products over the three months.

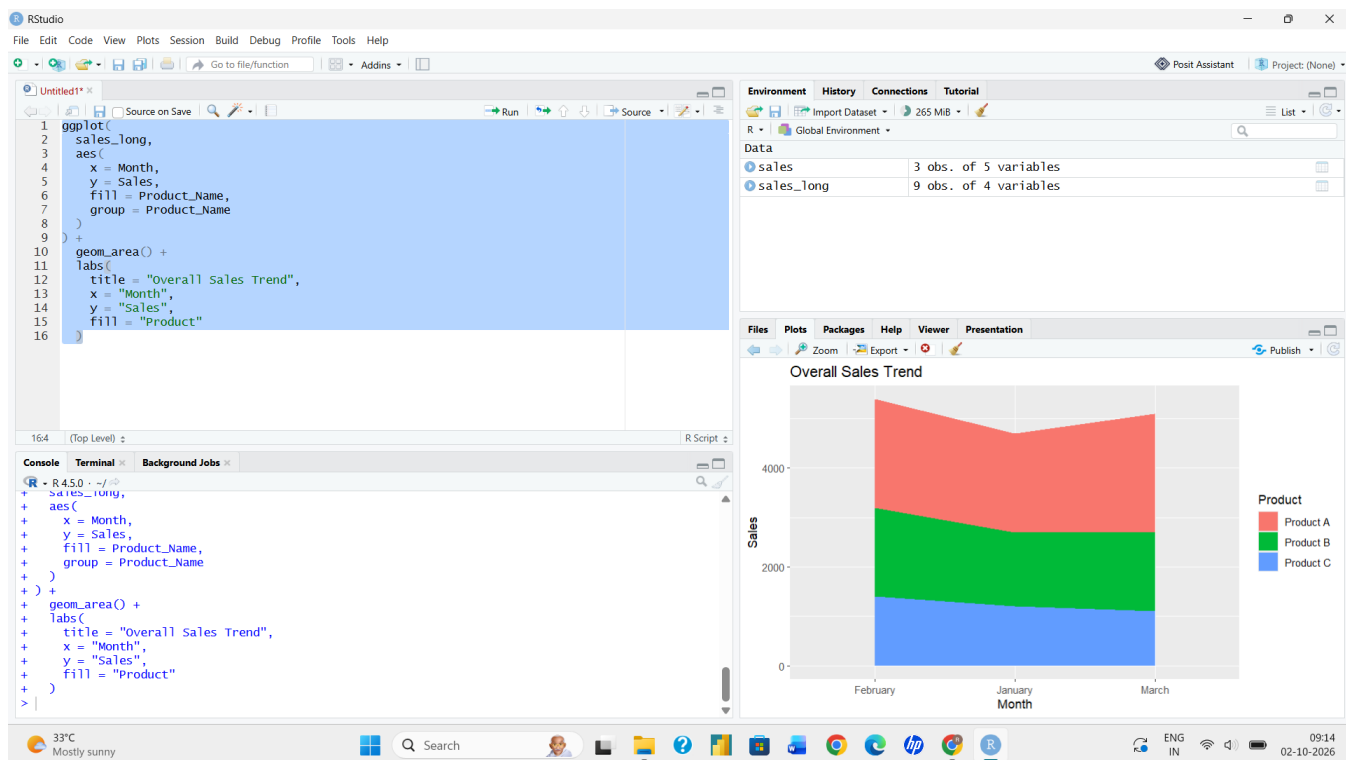
```
# Dataset
month <- c("January", "February", "March")
prodA <- c(2000, 2200, 2400)
prodB <- c(1500, 1800, 1600)
prodC <- c(1200, 1400, 1100)

# Cumulative values for stacking
c1 <- prodA
c2 <- prodA + prodB
c3 <- prodA + prodB + prodC
x <- 1:3

# Stacked Area Chart
plot(x, c3,
     type = "n",
     xaxt = "n",
     xlab = "Month",
     ylab = "Sales",
     main = "Overall Sales Trend of All Products",
     ylim = c(0, max(c3) + 1500))

polygon(c(x, rev(x)), c(c3, rev(c2)), col = "seagreen", border = NA)
polygon(c(x, rev(x)), c(c2, rev(c1)), col = "orange", border = NA)
polygon(c(x, rev(x)), c(c1, rep(0, 3)), col = "steelblue", border = NA)

axis(1, at = x, labels = month)
legend("topleft",
     legend = c("Product C", "Product B", "Product A"),
     fill = c("seagreen", "orange", "steelblue"))
```



Output 2: Stacked Area Chart of Overall Sales

Observation: Total sales rise from 4700 in January to 5400 in February and then fall to 5100 in March. Product A contributes the largest share of the total in every month.

PROGRAM 3: Table of Monthly Sales Data

Build a table to show the monthly sales data for each product. Label the table elements.

```
# Dataset
sales_table <- data.frame(
  Product_ID   = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  January_Sales = c(2000, 1500, 1200),
  February_Sales = c(2200, 1800, 1400),
  March_Sales   = c(2400, 1600, 1100)
)

# Monthly sales table
print(sales_table)
```

Console output:

	Product_ID	Product_Name	January_Sales	February_Sales	March_Sales
1	1	Product A	2000	2200	2400
2	2	Product B	1500	1800	1600
3	3	Product C	1200	1400	1100

Output 3: Monthly Sales Table

```
1 sales_long
```

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Console Terminal × Background Jobs ×

R 4.5.0 · ~/

```
2 Product B 1500 1800 1600
3 Product C 1200 1400 1100
```

sales_long

A tibble: 9 × 4

Product_ID	Product_Name	Month	Sales
<dbl>	<chr>	<chr>	<dbl>
1	Product A	January	2000
1	Product A	February	2200
1	Product A	March	2400
2	Product B	January	1500
2	Product B	February	1800
2	Product B	March	1600
3	Product C	January	1200
3	Product C	February	1400
3	Product C	March	1100

Observation: The table lists the January, February and March sales of every product, with each column labelled by product ID, product name and month.

PROGRAM 4: Interactive Product Sales Dashboard

Develop a dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data using the Shiny package.

```
library(shiny)

product <- c("Product A", "Product B", "Product C")
jan <- c(2000, 1500, 1200)
feb <- c(2200, 1800, 1400)
mar <- c(2400, 1600, 1100)

sales <- rbind(jan, feb, mar)
colnames(sales) <- product
rownames(sales) <- c("January", "February", "March")

sales_table <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = product,
  January_Sales = jan,
  February_Sales = feb,
  March_Sales = mar
)

ui <- fluidPage(

  titlePanel("Monthly Product Sales Dashboard"),

  tabsetPanel(
    tabPanel("Grouped Bar Chart", plotOutput("groupedPlot")),
    tabPanel("Stacked Area Chart", plotOutput("areaPlot")),
    tabPanel("Sales Table", tableOutput("salesTable"))
  )
)

server <- function(input, output) {

  output$groupedPlot <- renderPlot({
    barplot(sales,
      beside = TRUE,
      xlab = "Product",
      ylab = "Sales",
      main = "Product Sales in the First Quarter",
      col = c("skyblue", "orange", "seagreen"),
      legend.text = rownames(sales))
  })

  output$areaPlot <- renderPlot({
    x <- 1:3
    # Stack the three products across the three months
    pA <- sales[, 1]; pB <- sales[, 2]; pC <- sales[, 3]
    s1 <- pA; s2 <- pA + pB; s3 <- pA + pB + pC

    plot(x, s3, type = "n", xaxt = "n",
      xlab = "Month", ylab = "Sales",
      main = "Overall Sales Trend of All Products",
      ylim = c(0, max(s3) + 1500))
    polygon(c(x, rev(x)), c(s3, rev(s2)), col = "seagreen", border = NA)
```

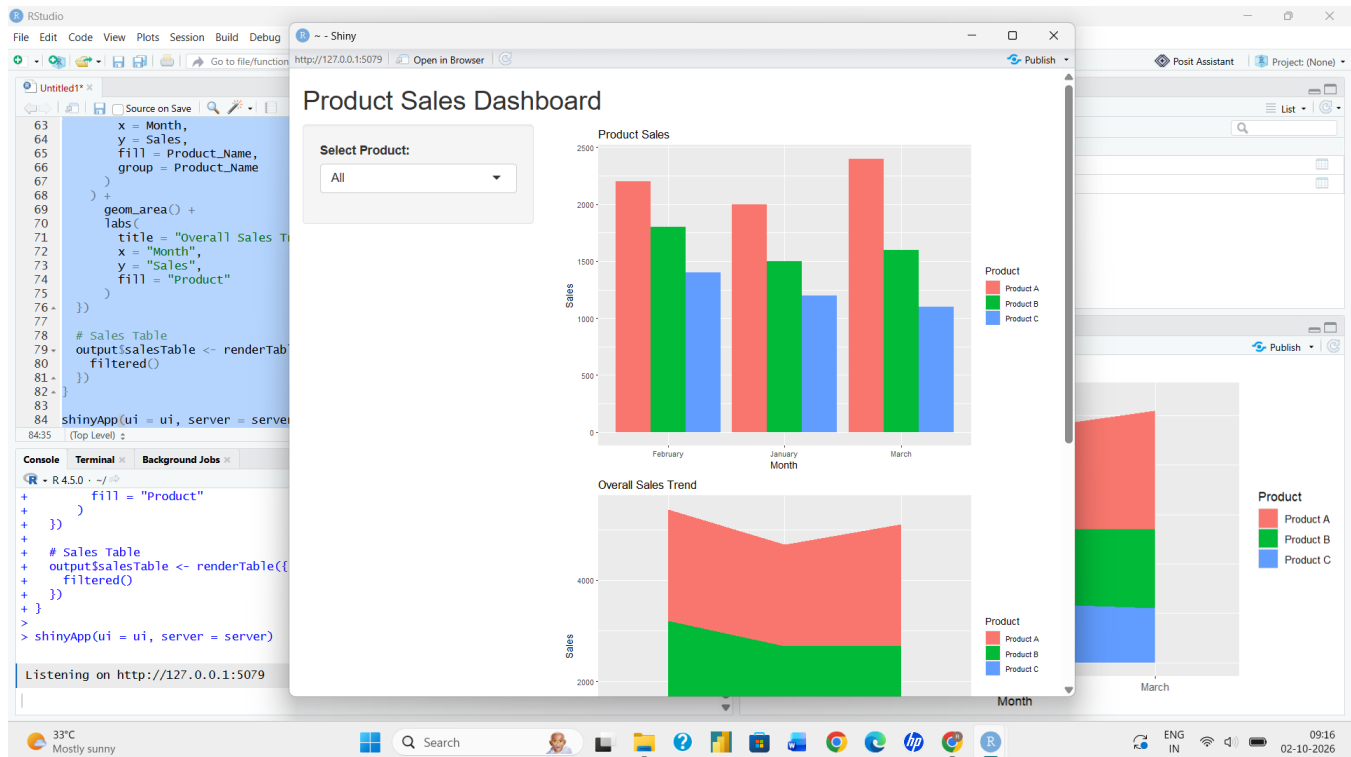
```

polygon(c(x, rev(x)), c(s2, rev(s1)), col = "orange", border = NA)
polygon(c(x, rev(x)), c(s1, rep(0, 3)), col = "steelblue", border = NA)
axis(1, at = x, labels = rownames(sales))
legend("topleft", legend = c("Product C", "Product B", "Product A"),
      fill = c("seagreen", "orange", "steelblue"))
})

output$SalesTable <- renderTable({
  sales_table
})
}

shinyApp(ui = ui, server = server)

```



Output 4: Interactive Product Sales Dashboard

Observation: The dashboard has three tabs. Selecting Grouped Bar Chart, Stacked Area Chart or Sales Table displays the corresponding view of the quarterly sales data.

Procedure

1. Create the product and monthly sales vectors in R and combine them into a matrix.
2. Use `barplot()` with `beside = TRUE` to create the grouped bar chart.
3. Calculate the cumulative sales of the products and use `polygon()` to draw the stacked area chart.
4. Create a data frame with the product and monthly sales columns and display it using `print()`.
5. Install and load the Shiny package, then create the UI with `tabsetPanel()` and the server components for the dashboard.
6. Run the Shiny application and switch between the Grouped Bar Chart, Stacked Area Chart and Sales Table tabs.

Result

The monthly product sales data was successfully visualized using a grouped bar chart, stacked area chart, and table. An interactive Shiny dashboard was also created to explore all three views.

Key R Functions Used

Function	Purpose
<code>barplot()</code>	Creates grouped bar charts (<code>beside = TRUE</code>).
<code>polygon()</code>	Draws the filled areas of the stacked area chart.
<code>data.frame()</code>	Creates the sales table.
<code>print()</code>	Displays the table in the console.
<code>tabsetPanel()</code> / <code>tabPanel()</code>	Create the tabbed dashboard layout.
<code>renderPlot()</code> / <code>renderTable()</code>	Generate plots and tables dynamically in Shiny.
<code>shinyApp()</code>	Runs the interactive Shiny application.